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PROJECT WORK ON PREDICTION OF DIABETES
AMONG FEMALES ON DIFFERENT HEALTH
PARAMETERS USING LOGISTIC REGRESSION

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INTRODUCTION

What is diabetes?

Diabetes Mellitus is a physical condition in which the pancreas does not produce enough insulin. Insulin is a hormone which controls the amount of glucose level by converting the excess glucose in the blood into glycogen. If the body lacks insulin, the excess glucose remains in the blood thereby increasing the blood sugar level and resulting in diabetes mellitus. Doctors often use the full name diabetes mellitus instead of only diabetes alone, in order to distinguish this disorder from diabetes insipidus which is a completely different disease.

India is known as the diabetes capital of the world with around 50 million people suffering from it. Diabetes is often called the silent killer since it kills the affected person slowly over the years. It can lead to kidney failure, atherosclerosis, neuropathy in the limbs and blindness. Frequent urination and thirst; loss of weight, blurry vision, dryness of the skin are the major symptoms which helps in suspecting diabetes mellitus.

However, doctors and medical researchers have said that, if the disease can be detected at an early stage followed by proper treatments, then it can go a long way in helping the patients live a normal life.

Diabetes in females

Between the year 1971 and 2000, the Annals of Internal Medicine had conducted a study where it stated that, the Death Rates for men with diabetes had fallen significantly. This decrease reflects the medical advancement in diabetes treatment. But at the same time, the survey also indicated that the Death Rates for women with diabetes did not improve much. The death rates in diabetes are higher among the women for the following reason:

- Diabetes increases the risk of heart disease by almost four times in women than in men. They are also at higher risk of other diabetes-related complications such as blindness, kidney, depression, etc.
- Some of the diabetes related complications are difficult to diagnose in women than in men
- Women often develop gestational diabetes during pregnancy which can often become dangerous.
- Hormones and inflammation act differently in women.

Though diabetes is a deadly disease, it often proves to be more fatal in women than in men. There is no permanent cure for this disease and once diagnosed with it, one can only manage its symptoms. In United States, an estimated 7.3 million adults ages 18 years or older have

diabetes but are undiagnosed (21.4 percent of adults with diabetes). Thus, the early detection of diabetes is crucial part in the diagnosis of diabetes.

SOURCE OF THE DATA

The dataset used in this project has been taken from the National Institute of Diabetes and Kidney Diseases.

EXPLORING THE DATA

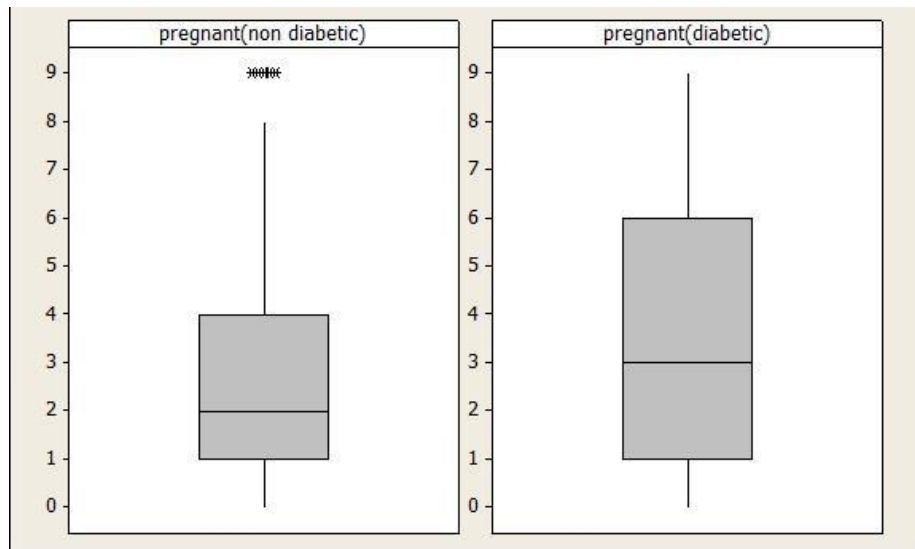
The dataset used in this project contains information of 376 women across 8 variables. The outcome tested was Diabetes, 115 of them tested positive and 261 of them tested negative. Therefore, there is one response (dependent) variable and 7 health parameters that could and should be related to the onset of diabetes and its future complications.

The binary response variable (denoted as Y) is defined as:

$$Y = \begin{cases} 1, & \text{the patient has diabetes} \\ 0, & \text{otherwise} \end{cases}$$

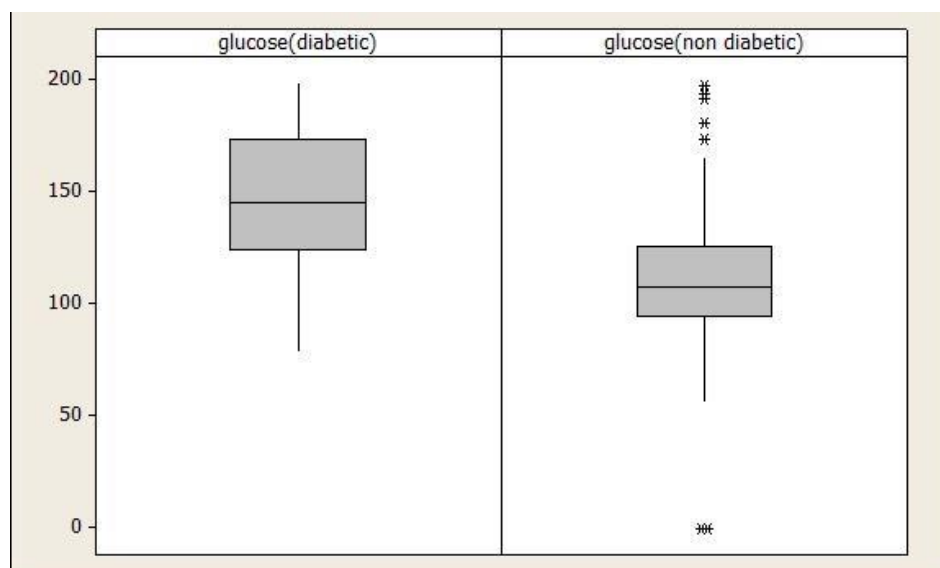
The 7 covariates are defined below:

1. **Pregnancies:** It displays the number of times the woman got pregnant. It is a discrete variable and is labelled as x_1 in our data. Gestational diabetes is a type of diabetes that can develop in a pregnant woman who don't already have diabetes. Every year, almost 10% of the pregnant women gets affected by the gestational diabetes. So, this covariate has been included in the dataset.



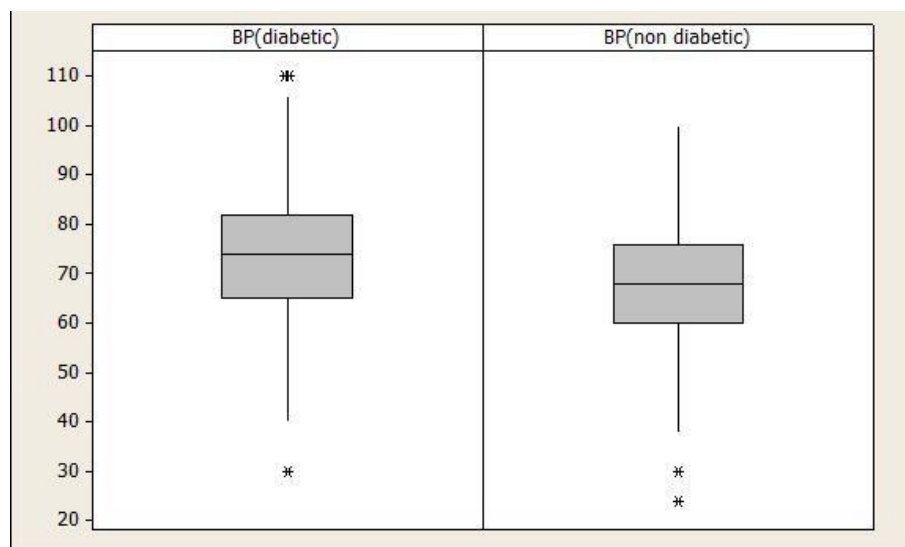
From the above graph, we can observe that, on an average the number of pregnancies in a diabetic female is slightly greater than that of a non-diabetic female. Thus, our data supports the fact that the chances of getting diabetes increase with the increase in the number of pregnancies for a female.

2. Glucose concentration: It displays the concentration of glucose in the blood (in mg/dL) and it is measured by 2 hours of oral glucose tolerance test (OGTT). It is a discrete variable. The normal fasting blood glucose level is lower than 110 mg/dL. A person having an OGTT greater than 145mg/dL is likely to have diabetes.



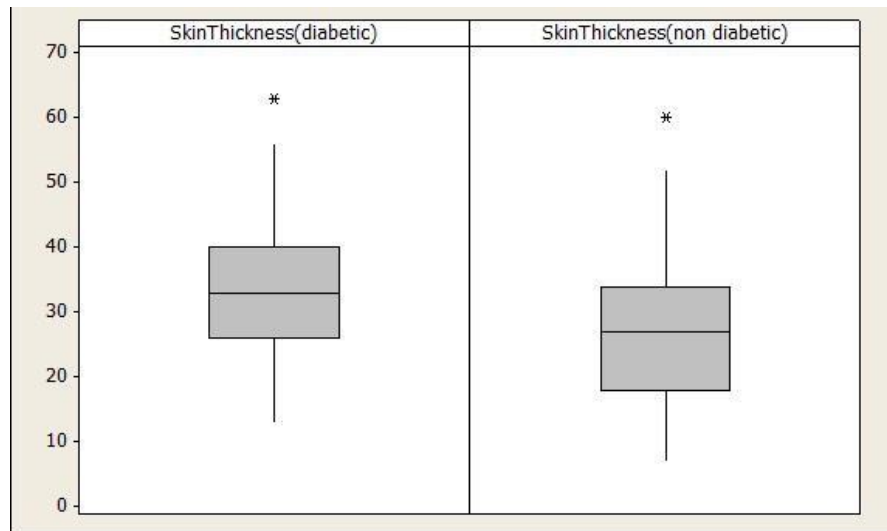
From the above graph, we can conclude that, on an average the diabetic person has a higher glucose concentration in their blood compared to that of non-diabetic person in our dataset.

3. Blood Pressure: It displays the diastolic blood pressure (in mm Hg) of the person. It is a discrete variable. The normal diastolic blood pressure is generally lesser than 80 mm Hg. Diabetes results in narrowing of the arteries and this condition is known as atherosclerosis. This causes High Blood Pressure, and it can often lead to heart attacks and renal failure. Thus, high blood pressure can be an indication of diabetes.



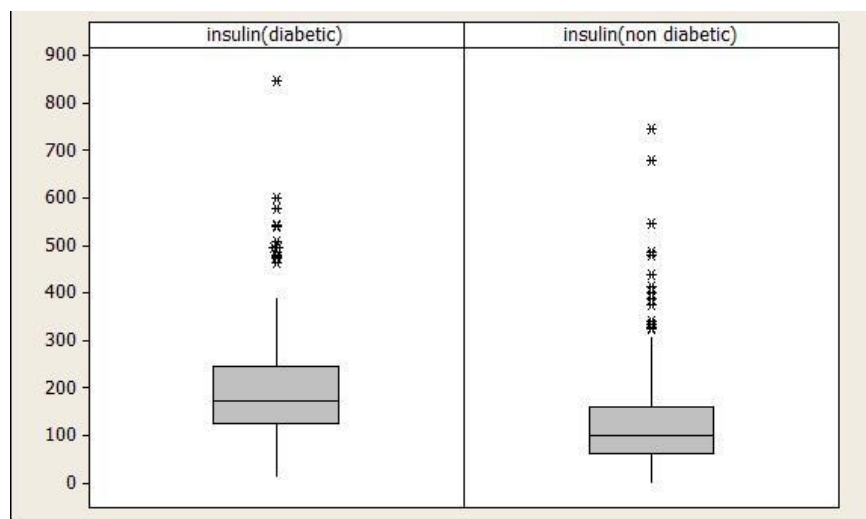
From the graph, we can say that the average Blood Pressure of diabetic females in the dataset is slightly more than that of non-diabetic females. Though there are out few outliers present in both the cases.

4. Skin thickness: It displays the triceps skin fold thickness (in mm). To be more precise, it is a measure of the obesity. It is a discrete variable. Almost 90% of the diabetic patients are obese. Overweight causes increased levels of fatty acids and inflammation which in turn leads to insulin resistance, thereby leading to diabetes mellitus.



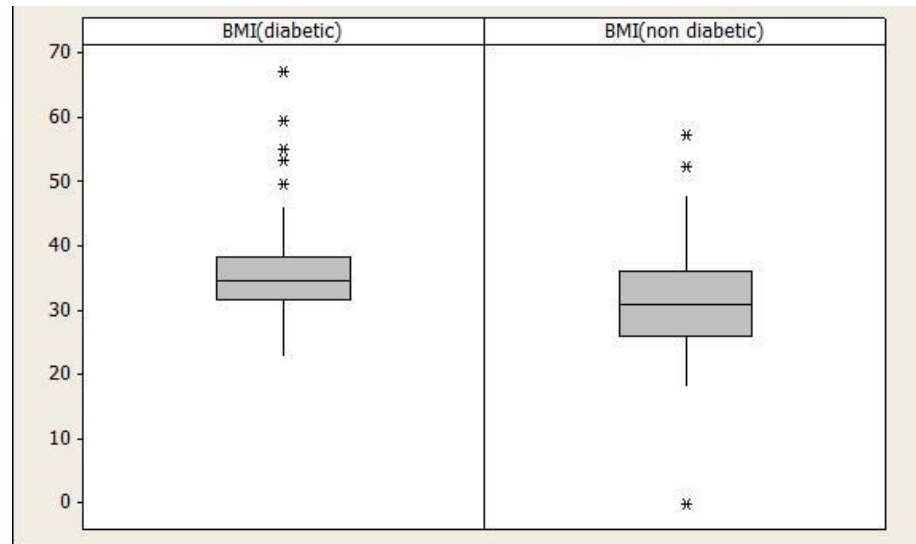
From the boxplot, we observe that the average triceps skinfold thickness is slightly more in diabetic females than the non-diabetic females in the dataset. Thus, our dataset supports the fact that obesity can be a reason for the occurrence of diabetes in females.

5. Insulin: It displays the amount of insulin concentration in the blood (in μ U/ml). The two-hour glucose tolerance test with Insulin levels is used to assess how an individual processes glucose and how the insulin in the body responds to those glucose levels.

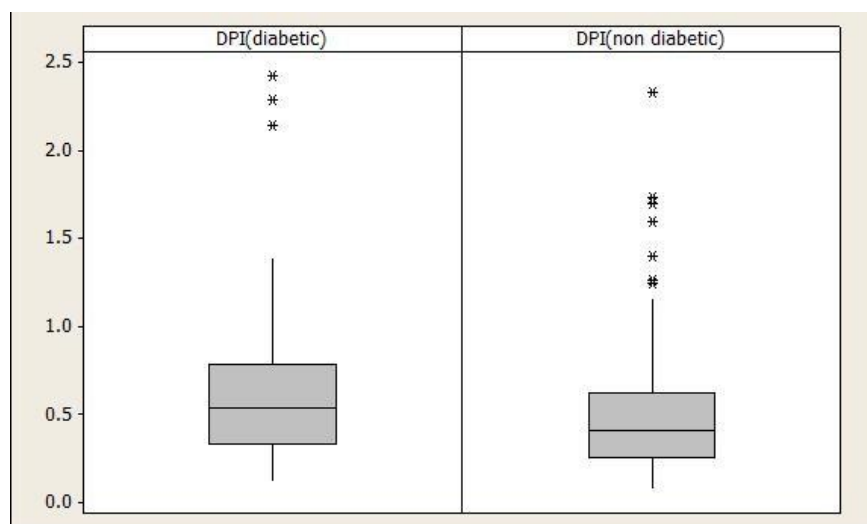


From the above boxplot, we can observe that the average insulin concentration in the blood for diabetic females is more than that of non-diabetic females in the dataset. Information obtained from the dataset is quite unexpected because diabetes mellitus is the condition in which the body fails to produce much insulin. As a result, a diabetic patient lacks insulin in their body. However, we can also see several outliers present in the dataset.

6. Body Mass Index (BMI): It displays the body mass index of the person (in Kg/m^2). BMI is calculated as the weight of the person (in kg) divided by their height in (metre sq.). Therefore, a person who is overweight has a high BMI and a person who is underweight has low BMI.



7. Diabetes pedigree function: it is a function which scores the likelihood of diabetes based on the family history. It provided some data on diabetes mellitus history in relatives and the genetic relationship of those relatives to the person. This measure of genetic influence gave us an idea of the hereditary risk one might have with the onset of diabetes mellitus.



From the above plot, we can observe that Diabetes Pedigree Function (DPI) is slightly more for the diabetic females than that of non-diabetic females. The median DPI for the diabetic females is 0.553 whereas, the median DPI for the non-diabetic females is 0.405. Thus, our dataset supports the claim that a person with family history of diabetes is more likely to develop diabetes.

The first 12 rows of the dataset are given below:

	A	B	C	D	E	F	G	H
1	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Outcome
2	1	89	66	23	94	28.1	0.167	0
3	0	137	40	35	168	43.1	2.288	1
4	3	78	50	32	88	31	0.248	1
5	2	197	70	45	543	30.5	0.158	1
6	1	189	60	23	846	30.1	0.398	1
7	5	166	72	19	175	25.8	0.587	1
8	0	118	84	47	230	45.8	0.551	1
9	1	103	30	38	83	43.3	0.183	0
10	1	115	70	30	96	34.6	0.529	1
11	3	126	88	41	235	39.3	0.704	0
12	1	97	66	15	140	23.2	0.487	0

OBJECTIVE

Diabetes is one of the most common diseases around the world and if detected earlier, it may prevent the progression of the disease and other complications. Thus, the main aim of this project is to help in the early detection of diabetes. Our primary objective is to identify which health parameters provided in the dataset plays the most significant role in the occurrence of diabetes in females. After identifying the significant covariates, now we can use suitable statistical tools to predict whether a woman has a potential chance of developing diabetes or not based on the information provided on the dataset.

MODEL SELECTION

Here we take OUTCOME as our response variable (y , say) and other variables as our predictor variable ($x_1, x_2, \dots, x_7$, respectively say) where (x_1) is indicator variable and other as continuous variable.

Here we assume that y is stochastic random variable and $x_1, x_2, \dots, x_7$ as our nonstochastic random variable. y is binary random variable taking the value 1 with probability p (say) and 0 with probability $1-p$. Therefore, $E(y) = 1 \times p + 0 \times (1-p)$

$= p$

Now at first we take a multiple linear regression model such as,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_7 x_7 + e.$$

Where e is error.

Now, from the above model we get $E(y)$

$$= E(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_7 x_7 + e)$$

$$= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_7 x_7 \text{ (since we assume } E(e)=0, \text{ which is reasonable assumption)}$$

$$\text{Therefore from this two we get } p(x_1, x_2, \dots, x_7) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_7 x_7$$

Now, the R.H.S. is always lies between 0 to 1 but L.H.S. can take any value in the real line as it is a linear

function of some continuous and indicator variable. So, in many times predicted value $p(x_1, x_2, \dots, x_7)$ is less than 0 or greater than 1. Therefore, the model fails to fit data adequately.

So, we use the generalized linear model for binary data.

Here we write, $y = \mu + e \dots\dots\dots(1)$ where $\mu = E(y)$, since $E(e) = 0$.

Here the function $g(\cdot)$ that relates μ to the linear predictor as

$g(\mu) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_7 x_7$, $g(\cdot)$ is called the link function.

The R.H.S is called the systematic components and the y in (1) is called the random component and $g(\cdot)$ connects the random and systematic components.

Here, we take the $g(\mu)$ as $g(\mu) = \log[\mu/(1 - \mu)]$ models the log of odds. It is appropriate when μ is between 0 and 1, such as a probability. This is called the logit link. A GLM that uses the logit link is called a logistic regression model.

Now, $\mu = p(x_1, x_2, \dots, x_7) = \frac{e^{\beta_1 x_1 + \beta_2 x_2 + \dots + \beta_7 x_7}}{1 + e^{\beta_1 x_1 + \beta_2 x_2 + \dots + \beta_7 x_7}}$,

in x_1 .

Similarly, $\beta_2, \beta_3, \dots, \beta_7$ can be interpreted as similarly.

MODEL FITTING AND IMPROVISATION

The r code and outcomes for first model is given below.

```
library(readxl)
syantanexcel <- read_excel("D:/PYTHON/syantanexcel.xlsx", sheet = "syantanexcel")
View(syantanexcel)
model_1=glm(formula=Outcome~Pregnancies+Glucose+BloodPressure+SkinThickness+Insulin+BMI+DiabetesPedigreeFunction,syantanexcel,family="binomial")
summary(model_1)
```

```
Call:
glm(formula = Outcome ~ Pregnancies + Glucose + BloodPressure +
    SkinThickness + Insulin + BMI + DiabetesPedigreeFunction,
    family = "binomial", data = syantanexcel)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.5732  -0.7359  -0.4230   0.7425   2.9851

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -8.1736736  0.6963139  -11.738  < 2e-16 ***
Pregnancies    0.1484580  0.0281044   5.282 1.28e-07 ***
Glucose        0.0363516  0.0036544   9.947  < 2e-16 ***
BloodPressure  -0.0118207  0.0051409  -2.299  0.02148 *
SkinThickness  -0.0003750  0.0068389  -0.055  0.95628
Insulin       -0.0012783  0.0008963  -1.426  0.15383
BMI            0.0880609  0.0150195   5.863 4.54e-09 ***
DiabetesPedigreeFunction 0.9675342  0.2984012   3.242  0.00119 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 993.48  on 767  degrees of freedom
Residual deviance: 725.97  on 760  degrees of freedom
AIC: 741.97

Number of Fisher Scoring iterations: 5
```

By fitting the model we can see that Skin Thickness and insulin are not contributing significantly towards the prediction of the female is diabetic or not. So here we drop the less contributing factors and comparing the AIC value and residual deviance consequently. And by doing those dropping of significant factors we can find that by dropping skin thickness, we can optimize the model significantly. The R code and outcomes for optimized model is given below.

```
model_2=glm(formula=Outcome~Pregnancies+Glucose+BloodPressure+Insulin+BMI+DiabetesPedigreeFunction+syantanexcel,family="binomial")
summary(model_2)
```

```
Call:
glm(formula = Outcome ~ Pregnancies + Glucose + BloodPressure +
    Insulin + BMI + DiabetesPedigreeFunction, family = "binomial",
    data = syantanexcel)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.5701  -0.7352  -0.4230   0.7420   2.9853

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    -8.1726681   0.6960005  -11.742 < 2e-16 ***
Pregnancies     0.1485506   0.0280578   5.294 1.19e-07 ***
Glucose         0.0363867   0.0035990  10.110 < 2e-16 ***
BloodPressure  -0.0118662   0.0050730  -2.339 0.01933 *
Insulin        -0.0012997   0.0008074  -1.610 0.10745
BMI             0.0878198   0.0143574   6.117 9.55e-10 ***
DiabetesPedigreeFunction 0.9661399   0.2972629   3.250 0.00115 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 993.48  on 767  degrees of freedom
Residual deviance: 725.97  on 761  degrees of freedom
AIC: 739.97

Number of Fisher Scoring iterations: 5
```

So the optimized fitted model is-

= -8.1726681+0.1485506*Pregnancies+0.0363867*Glucose-0.0118662*Blood Pressure-0.0012997*insulin+0.0878198*Diabetes Pedigree Function

CONCLUSION

From the above study by fitting a generalized linear model, we can conclude that 'the no. of pregnancies', 'Glucose level in model', 'Body Mass Index', 'Blood Pressure', 'Insulin', 'Diabetes Pedigree Function' are the most significant factors in predicting the presence of diabetes in females. By observing the coefficients of the predictor variables, we conclude that increase in the glucose level, greater number of pregnancies, low levels of insulin and the family history can increase the risk of diabetes in a female.

DISCUSSIONS

Further work can be done on this project by using ANOVA to find the significant predictor variables to predict a female is diabetic or not. Also we can use to multicollinearity concept to find of if a subset of this given set of covariates is sufficient enough to to predict whether a female has a high chance of getting diabetes or not.

REFERENCE

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- McCullagh, P & Nelder, J.A.(1995), Generalized Linear Models. Chapman and Hall.
- www.medium.com

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